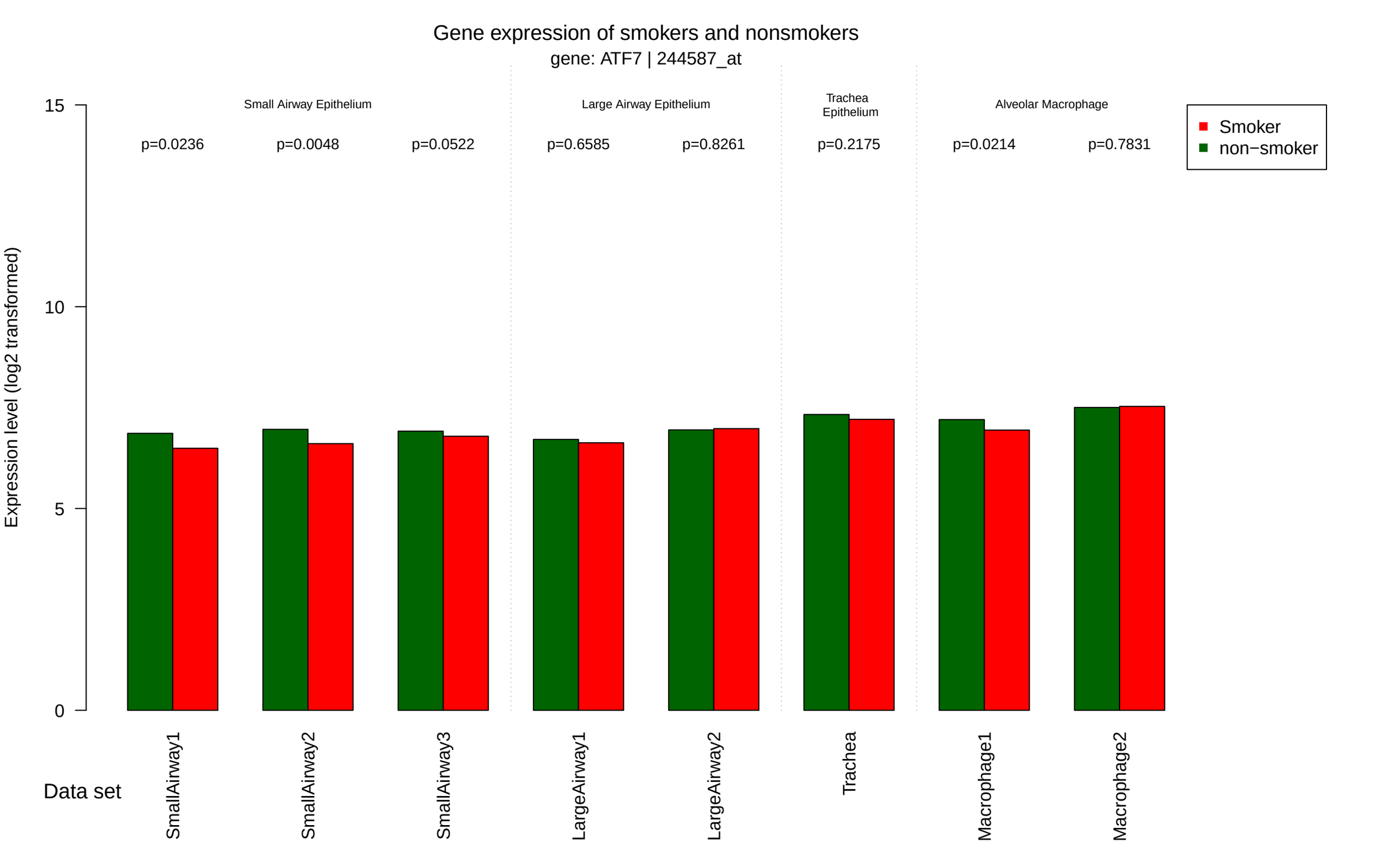
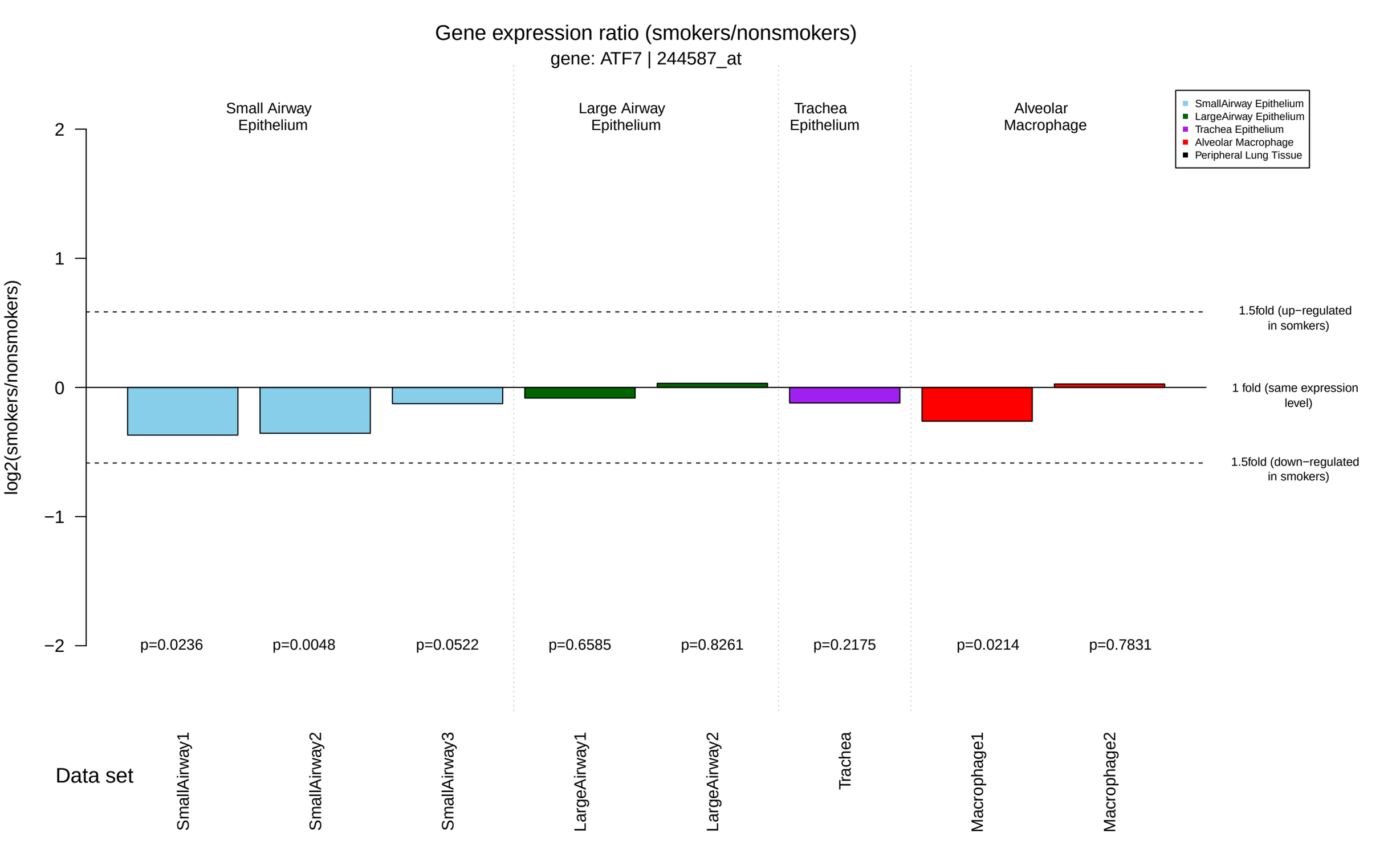
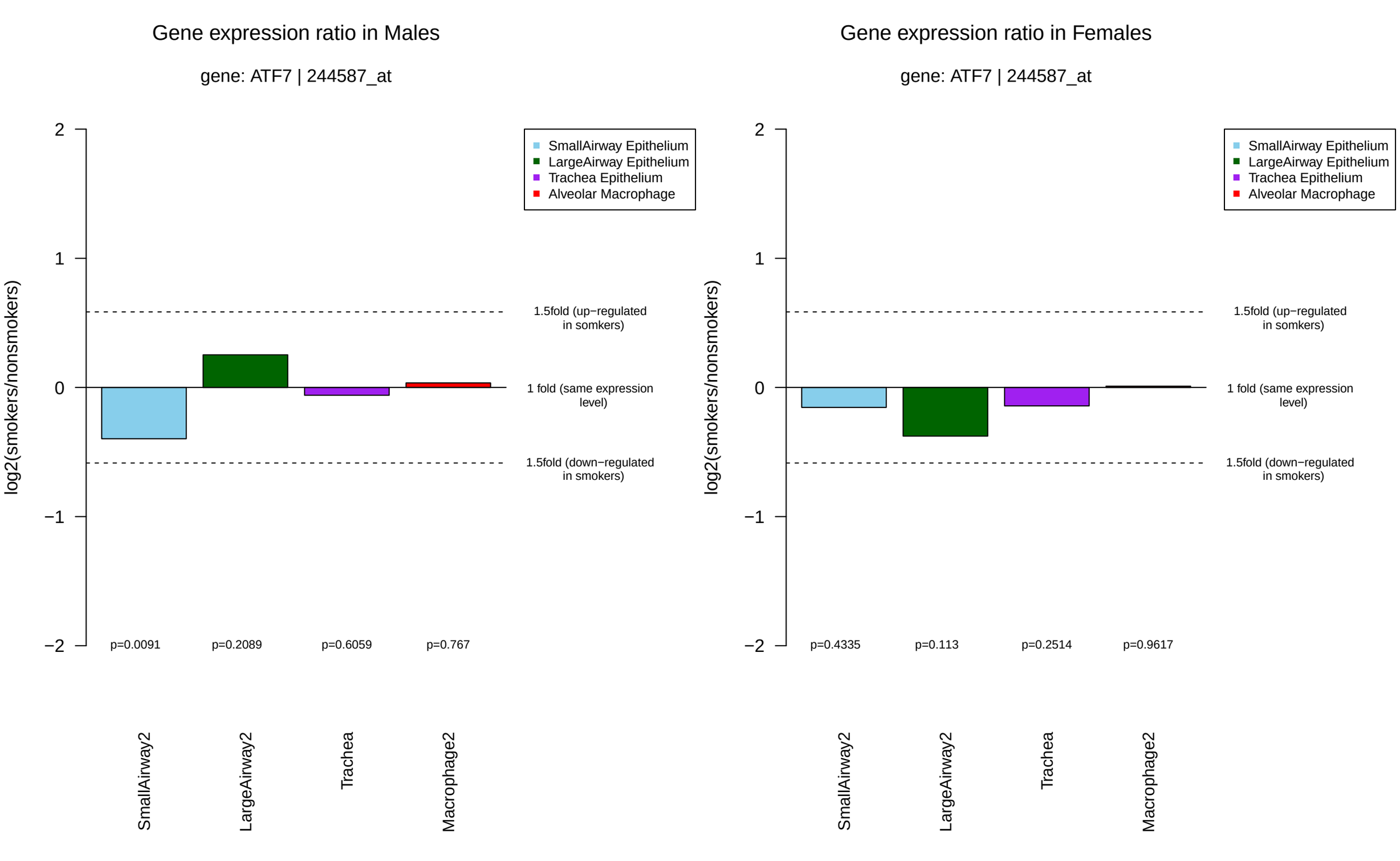
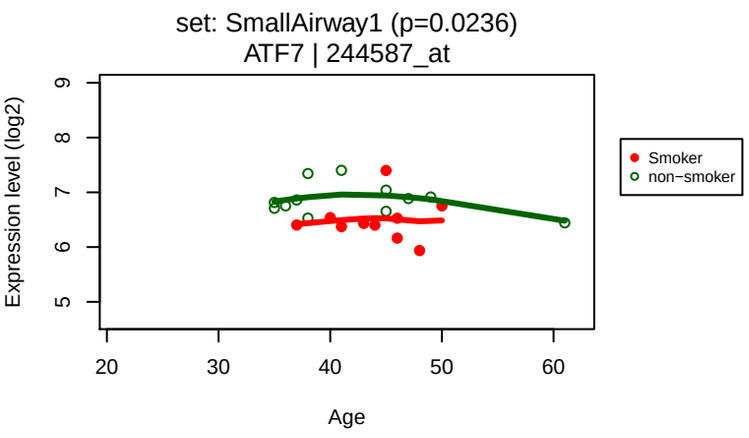
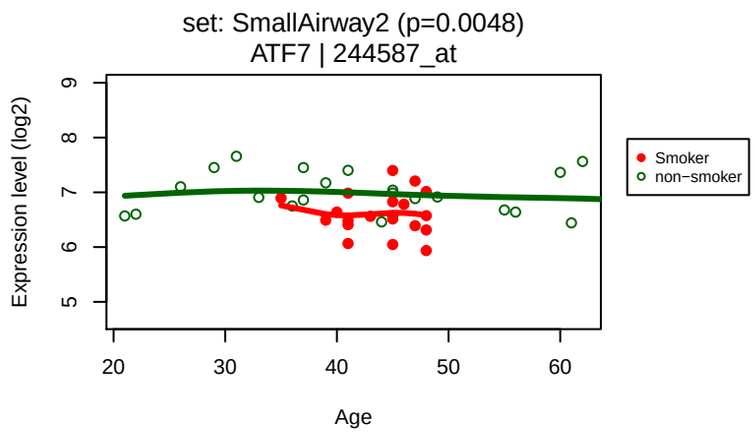
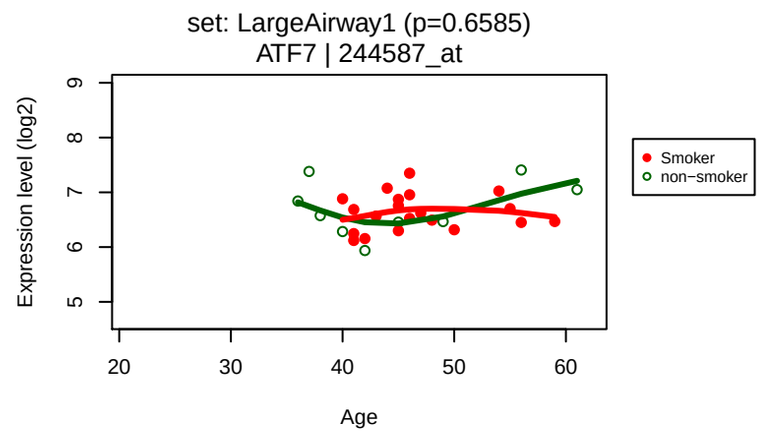

Gene expression of smokers and nonsmokers
gene: ATF7 | 244587_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0236
p=0.0048
p=0.0522
p=0.6585
p=0.8261
p=0.2175
p=0.0214
p=0.7831
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

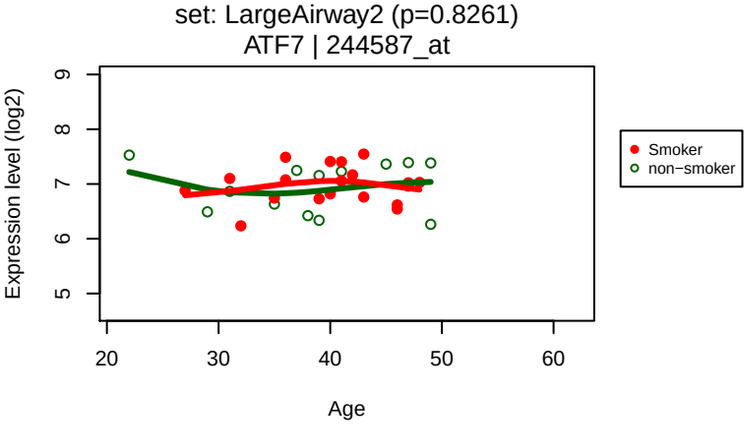

Gene expression ratio (smokers/nonsmokers)
gene: ATF7 | 244587_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0236
p=0.0048
p=0.0522
p=0.6585
p=0.8261
p=0.2175
p=0.0214
p=0.7831
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

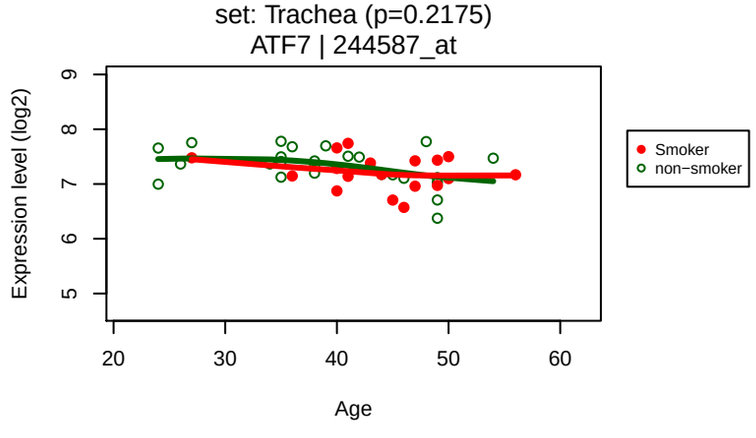

Gene expression ratio in Males
gene: ATF7 | 244587_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0091
p=0.2089
p=0.6059
p=0.767
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATF7 | 244587_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4335
p=0.113
p=0.2514
p=0.9617
SmallAirway2
LargeAirway2
Trachea
Macrophage2

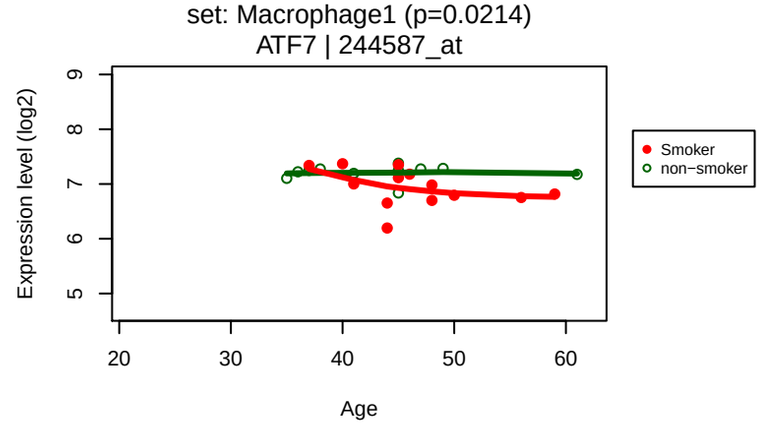

set: SmallAirway1 (p=0.0236)
ATF7 | 244587_at
Expression level (log2)
Age
Smoker
non−smoker

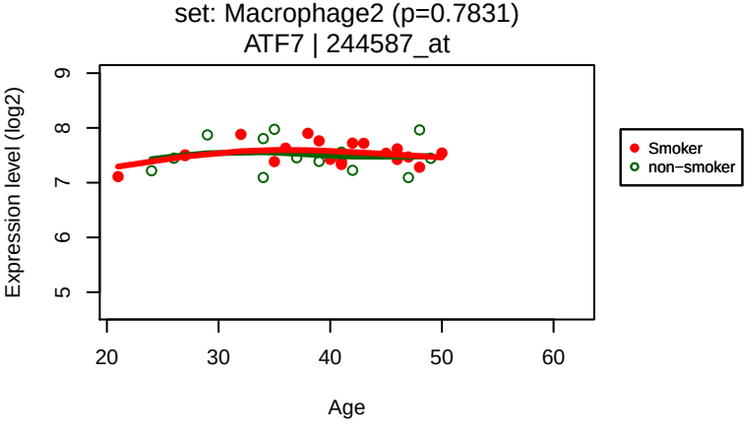

set: SmallAirway2 (p=0.0048)
ATF7 | 244587_at
Expression level (log2)
Age
Smoker
non−smoker

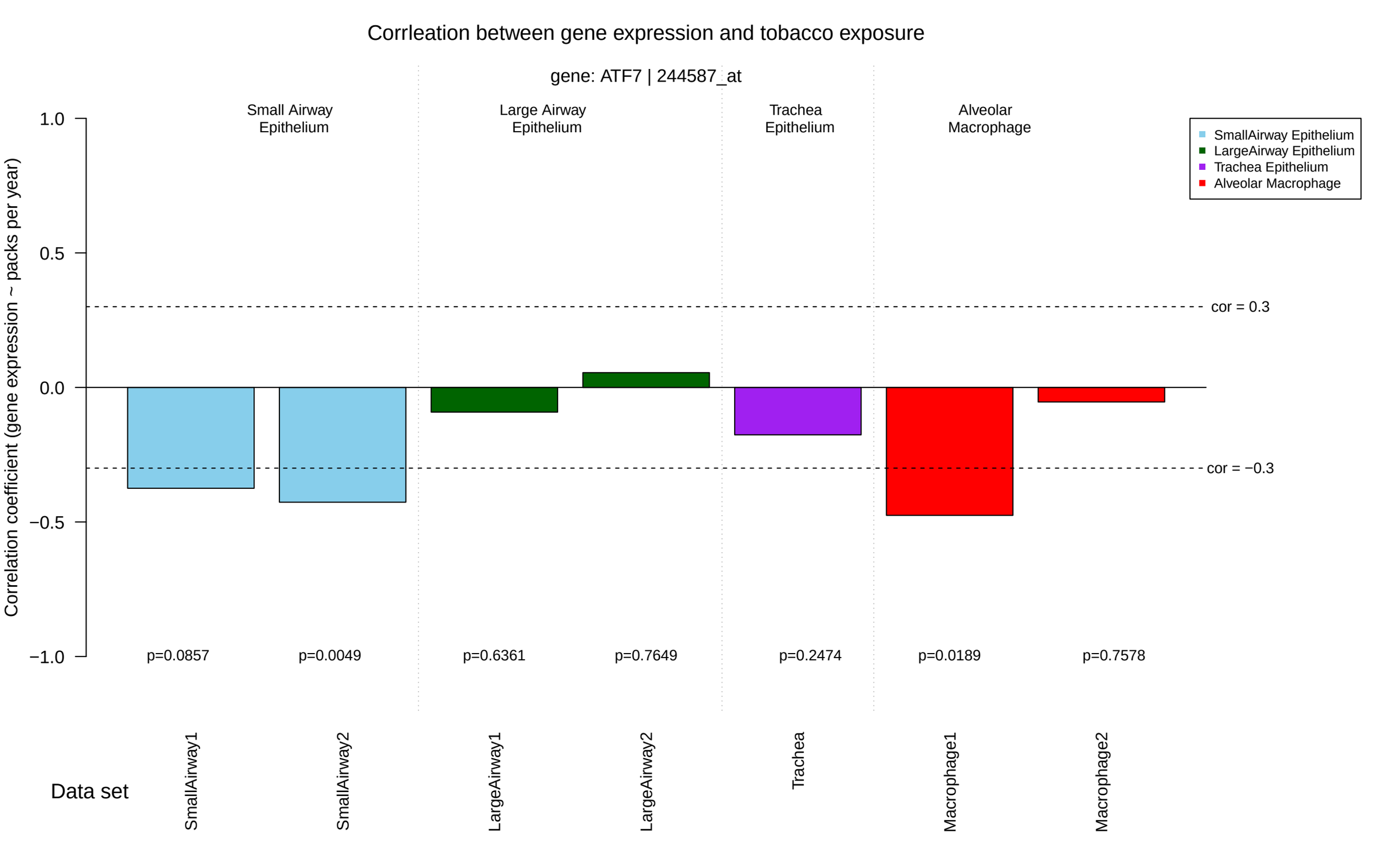

set: LargeAirway1 (p=0.6585)
ATF7 | 244587_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8261)
ATF7 | 244587_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2175)
ATF7 | 244587_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0214)
ATF7 | 244587_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7831)
ATF7 | 244587_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATF7 | 244587_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0857
p=0.0049
p=0.6361
p=0.7649
p=0.2474
p=0.0189
p=0.7578
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2